

## Example #02.

Prove that Square of an even integer is even.

Proof :-

Let  $m$  is an even integer.  
According to the definition,

$$m = 2K$$

for some integer  $K$ .

$$m^2 = (2K)^2$$

$$m^2 = 4K^2$$

$$m^2 = 2(2K^2)$$

$$\text{Put } 2K^2 = \ell$$

$$m^2 = 2\ell$$

Hence,  $m^2$  is also an even integer.

## Example # 04.

Prove that sum of any three consecutive integer is divisible by 3.

Proof :-

Let  $K, K+1, K+2$  be any three consecutive integers.

$$(K) + (K+1) + (K+2) = K + K+1 + K+2$$

$$(K) + (K+1) + (K+2) = 3K+3$$

$$(K) + (K+1) + (K+2) = 3(K+1)$$

If we divide the above equation by 3.

then,

$$\frac{(K) + (K+1) + (K+2)}{3} = \frac{3(K+1)}{3}$$

$$\frac{(K) + (K+1) + (K+2)}{3} = K+1$$

Hence, Proved that sum of any three consecutive integers is divisible

# Exercise

## Example #01.

Prove that the product of an even integer and an odd integer is even.

Proof:-

Let  $m$  is an even integer  
So,

$$m = 2K \text{ for some integer } K.$$

Let  $n$  is an odd integer,

$$n = 2K+1 \text{ for some integer } K.$$

$$mn = 2K(2K+1)$$

$$mn = 2K(2K+1)$$

$$mn = 4K^2 + 2K$$

$$mn = 2(2K^2 + K)$$

$$\text{Put } Q = 2K^2 + K$$

$$mn = 2Q$$

Hence,  $mn$  is an even.

~~Hence~~ proved.

## Example #03.

Prove that if  $n$  is an odd integer, then  $n^3 + n$  is even.

Proof :-

Let  $n$  is an odd integer.

According to the definition,

$$n = 2k + 1$$

Now,

$$n^3 + n = (2k+1)^3 + (2k+1)$$

$$n^3 + n = 8k^3 + 6k(2k+1) + 1 + 2k+1$$

$$n^3 + n = 8k^3 + 12k^2 + 6k + 2k + 2$$

$$n^3 + n = 8k^3 + 12k^2 + 8k + 2$$

$$n^3 + n = 2(4k^3 + 6k^2 + 4k + 1)$$

$$\text{Put } \mathcal{Q} = 4k^3 + 6k^2 + 4k + 1$$

$$n^3 + n = 2\mathcal{Q}$$

Hence, proved that  $n^3 + n$  is even. 2022.09.13 10:03